Algebra 1	Name____________________
Unit 1	Date_____________________
Lesson 9 Homework	Period____________________

1. Which of the following expressions can be combined?

A. $-3\sqrt{25} - 25\sqrt{3}$
B. $\sqrt{7} + 4\sqrt{14}$
C. $2\sqrt{12} - 6\sqrt{27}$
D. $5\sqrt{25} + \sqrt{20}$

Three students simplified the expression $6\sqrt{27} - 5\sqrt{12}$ below. Their work is shown below.

Marc	**Jess**	**Tamara**
$6\sqrt{27} - 5\sqrt{12}$ $1\sqrt{13}$ $\sqrt{13}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $54\sqrt{3} - 20\sqrt{3}$ $34\sqrt{3}$	$6\sqrt{27} - 5\sqrt{12}$ $6\sqrt{9 \cdot 3} - 5\sqrt{4 \cdot 3}$ $18\sqrt{3} - 10\sqrt{3}$ $8\sqrt{3}$

2. Which student simplified the expression correctly?

3. Write a note to one of the students who made an error explaining their mistake.

4. Write a note to the other student who made an error explaining their mistake.

For question 5-10, find the sum or difference of the expression. If you can not add or subtract the expression, explain why.

5. $-2\sqrt{18} + \sqrt{50}$

6. $\sqrt[3]{56} - \sqrt[3]{189}$

7. $6\sqrt{8} - 8\sqrt{6}$

8. $200^{\frac{1}{2}} - 1000^{\frac{1}{3}}$

9. $-\sqrt{80} + 6\sqrt{5} - 3\sqrt{20}$

10. $\sqrt[3]{250} - 4\sqrt[3]{54}$